

MIT 805: Big Data Semester Project (2026)

Large-Scale Data Analysis and Distributed Processing with PySpark

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1. Introduction

The growth of global populations, rapid technological advances, and expanding industries produces massive amounts of data daily. This assignment provides a structured, hands-on approach to dealing with Big Data, focusing on real-world domains such as Medical, Transport, and Stock Exchange. Working in **groups of two**, you will learn to collect, process, reduce, and visualize data, while also critically analyzing it through the 7 Vs of Big Data and identifying business value.

2. Group Allocation

- This is a **group project**, completed in **pairs (2 members per group)**.
- Groups have been **pre-allocated on the ClickUP LMS**; check your group partner there before proceeding.
- Both members are jointly responsible for all deliverables; GitHub commit history will be used to verify individual contribution.

3. Learning Outcomes

By completing this project, you will be able to:

- Collect, process, and analyze real-world datasets collaboratively
- Implement MapReduce-style distributed data processing using PySpark and critically compare Spark's DAG-based execution model with traditional Hadoop MapReduce.
- Demonstrate the insights from the processing/analysis using visualization tools.
- Critically evaluate the business or societal value of your analysis.
- Work with professional tools (GitHub, LaTeX/Overleaf, Google Colab, PySpark).

- Evaluate the computational and data-management challenges associated with large-scale datasets.

4. Dataset Selection

Each group must choose a large dataset that is publicly available. Some examples include:

- **Medical:** (e.g. UCI Repository Healthcare, or other openly available medical data)
- **Transport:** (e.g. NYC Open Data – Transportation, or an alternative source)
- **Stock Exchange:** Kaggle Stock Market Datasets, Yahoo Finance

Dataset size: The **raw dataset** (original publicly available dataset) must fall between **25GB and 40GB**. The working dataset (the portion downloaded/preprocessed for the project) should not be less than 12GB, and the processing dataset (the data actually processed by PySpark) should not be less than 3 GB. You should report all 3 dataset sizes.

For the MapReduce implementation (Part 2), groups may use a representative subset of the full dataset that is large enough to demonstrate meaningful processing and results, but manageable within your available compute resources (e.g. Google Colab). Your report must clearly state the full raw dataset size and the size of the subset actually processed, with justification for the subset chosen.

5. Technical Implementation: PySpark in place of Hadoop

Implement your MapReduce-style processing using **PySpark**, rather than native Hadoop MapReduce. This substitution is permitted because PySpark is a widely used framework for large-scale distributed data processing and provides a practical industry-relevant alternative to implementing native Hadoop MapReduce. PySpark provides a practical way to implement MapReduce-style distributed processing without requiring students to configure a complex Hadoop cluster. Although PySpark is not identical to Hadoop MapReduce, its transformations and actions allow students to implement key distributed concepts, including mapping, shuffling, grouping, reduction, and aggregation; using both low-level RDDs and modern DataFrame APIs. Furthermore, Spark uses a Directed Acyclic Graph (DAG) execution model to optimize multi-stage computations in-memory, avoiding the heavy disk I/O bottlenecks of the traditional MapReduce framework.

6. Assignment Structure

Part 1 – Data Collection & Analysis (Due 03-09-2026)

1. **Dataset provenance and description:** Collect and describe your group's dataset. Identify the dataset's license or terms of use and confirm that the dataset is appropriate for academic use.
2. **Dataset size and characteristics:** Provide details: source, size (raw, working, processing), format, data collection period, publication date, and/or last update date, where available.

3. **Data quality and preparation:** Discuss how you prepared the data for analysis, and potential quality concerns such as missing data, data quality, temporal limitations, representativeness, computational constraints, or potential misleading correlations.
4. **EDA:** Carryout a thorough Exploratory Data Analysis of the dataset. You can use Python and existing libraries.
5. **Analyze and discuss** the dataset using **7 Vs** of Big Data:
 - Core 3 Vs: Volume, Velocity, Variety
 - Value (business/societal impact)
 - At least 3 others.
 - For each of the 7 Vs, you must provide evidence from your chosen dataset and explain why that characteristic matters. Generic definitions of the Vs without application to the dataset will receive limited credit.
6. Discuss **potential business/societal value and insights** (e.g., improving patient outcomes, optimizing transport routes, predicting market movements, financial value, operational efficiency, social outcomes, public policy, public health, etc.). Discuss at least two limitations of the dataset.

Deliverables:

- Written report (**PDF, maximum 2 pages**) – Upload to ClickUP. Submissions by email will not be graded. The 2-page limit applies to the main narrative. Supporting tables, figures, code outputs and additional visualizations may be included in an appendix.
- GitHub repo link with project setup/code. The structure of your Github repo should in the form given. Provide the link in your report.

```
project/
├── README.md
├── requirements.txt
├── data/
│   └── README.md
├── notebooks/
├── src/
├── output/
├── figures/
└── report/
```

Due Date: Thursday, 3rd September, 2026; 23:59

Part 2 – MapReduce & Visualization (Due 08-10-2026)

For Part 2, you are required to extract meaningful information using appropriate distributed processing operations, such as aggregation, filtering, grouping, joins, statistical summaries, correlations, or other techniques appropriate to the selected dataset. The PySpark implementation must demonstrate MapReduce-style distributed processing, including transformation/mapping, grouping or shuffle, and reduction/aggregation operations. Students must also provide evidence of Spark execution through an execution plan, Spark UI, DAG/stage information, or equivalent evidence. Students must distinguish Spark's DAG-based execution model from the traditional Map → Shuffle → Reduce model of Hadoop MapReduce. In summary, the substantive data processing must be performed using PySpark. Students should not convert the full processing dataset to Pandas or another single-machine framework for the main analysis. Pandas may only be used for small samples or final visualization where appropriate.

Your analysis should include the following:

1. **Processing objective:** Why are you analysing the data, what are you looking to achieve or obtain from such analysis. What is your central question?
2. **MapReduce-style implementation:** Process your dataset (or subset) using PySpark, implementing the MapReduce paradigm.
3. Spark transformations/actions
4. Shuffle/group/reduce operation
5. Distributed execution evidence
6. Results: Interpret results both technically and in terms of business value.
7. Visualization: Create visualizations (graphs, charts, dashboards).
8. Business/societal interpretation

Deliverables:

- Written report (**PDF, minimum 5 pages, maximum 7 pages, including visualizations**). Upload to ClickUP. You can include up to 2 additional appendix pages for more figures, only if needed. Note that your written report should document and explain your code/analysis as per the rubric, don't assume that it is in the repo, as it is assessed primarily in your written report; the Github repo is only as a reference/demonstration.
- GitHub repo with PySpark/MapReduce code. Provide the link in your report. **10-minute video demo** (analysis walkthrough, dataset, results, visualization, business insights). The **video link will be submitted via the Google Form** provided on ClickUP.

Extra Notes

1. You must use only publicly available datasets that permit educational/research use. Do not upload personally identifiable, confidential, or restricted health information to GitHub. Groups using medical datasets must document relevant privacy, ethical, licensing, and data-governance considerations.
2. GitHub Repository: Your git repository should ensure reproducibility. Another student should be able to understand how to reproduce the analysis from the README

without requiring undocumented steps. Do not upload datasets that are prohibited from redistribution. Provide the original dataset URL and instructions for obtaining it.

Due Date: Thursday, 8th October, 2026; 23:59

MARKING RUBRIC/GUIDELINES

PART 1 - Data Collection & Analysis: 18 Marks

Criterion	Marks	Marker guideline
1. Dataset source, licence & suitability	2	2: Reliable source, URL, licence/terms and academic suitability clearly established. 1: Source identified but licence/suitability incomplete. 0: Source/licensing not established.
2. Dataset characteristics & scale	3	3: Raw, working & processing sizes reported; format, records/variables, time period and relevant characteristics clearly described. 2: Most information provided. 1: Basic description only. 0: Required scale not demonstrated.
3. Data preparation & quality	3	3: Appropriate preparation plus meaningful discussion of missing data, duplicates, outliers, bias, representativeness and/or other quality issues. 2: Good cleaning/quality assessment but limited critical discussion. 1: Basic cleaning only. 0: Not demonstrated.

4. Exploratory Data Analysis (EDA)	4	4: Appropriate statistics/visualizations identify and interpret important patterns, relationships, trends or anomalies. 3: Good EDA with reasonable interpretation. 2: Basic EDA. 1: Mainly graphs/descriptive output with little interpretation. 0: No meaningful EDA.
5. 7 Vs of Big Data	4	4: Seven Vs clearly applied to the actual dataset with evidence and explanation of relevance. 3: Most Vs well applied. 2: Vs discussed but partly generic. 1: Mostly definitions with little dataset application. 0: Not addressed.
6. Value & limitations	2	2: Clear business/societal value plus at least two meaningful limitations. 1: Value or limitations discussed superficially. 0: Not demonstrated.

PART 1 TOTAL 18

PART 2 - PySpark, MapReduce & Analysis: 32 Marks

Criterion	Marks	What is assessed
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1. Processing objective & analytical question	4	4: Clear, focused analytical question; rationale; objectives; connection to the dataset and intended outcomes. 3: Clear question and rationale; minor gaps in focus or linkage. 2: Basic question/objective but somewhat broad or weakly justified. 1: Vague objective with little connection to the analysis. 0: No meaningful objective.
2. Mapping / Transformation	3	3: Appropriate transformation/mapping of data using PySpark; technically correct and relevant to the question. 2: Correct transformations but limited justification or minor technical issues. 1: Basic/superficial transformation. 0: Not demonstrated.
3. Shuffle / Grouping	3	3: Appropriate grouping/key-based operations; understanding of distributed data movement/shuffle. Meaningful grouping/shuffle correctly implemented; students explain why data must be grouped/moved. 2: Correct grouping with limited explanation of shuffle. 1: Basic grouping with little evidence of understanding. 0: Not demonstrated.
4. Reduction / Aggregation	4	4: Correct and meaningful reduction/aggregation; technically sound; directly produces information needed to answer the analytical question. 3: Correct aggregation with minor weaknesses.

		2: Basic aggregation but limited relevance or explanation. 1: Minimal/poorly justified aggregation. 0: Not demonstrated.
5. Spark execution & DAG	3	3: Provides Spark execution evidence and correctly explains DAG/stages/tasks/shuffle and Spark vs Hadoop MapReduce. 2: Good evidence but limited explanation. 1: Evidence provided with little interpretation. 0: No meaningful evidence.
6. Results & technical interpretation	3	3: Correct results; explains what the processing reveals and how it answers the analytical question. 2: Correct results with reasonable but limited interpretation. 1: Results presented with little interpretation. 0: No meaningful results.
7. Visualization	2	2: Appropriate, accurate and readable visualizations communicating key findings. 1: Basic or partly appropriate visualization. 0: Poor, misleading or absent visualization.
8. Business/societal interpretation & value	3	3: Findings clearly connected to stakeholders, practical applications/value and relevant limitations. 2: Reasonable interpretation and practical value identified. 1: General claims of usefulness with weak connection to findings.

		0: Not addressed.
9. Report writing & presentation	3	3: Clear, concise and professional report; logical structure; strong technical writing; effective figures/tables; appropriate referencing; within required length. 2: Generally well written and presented with some weaknesses. 1: Understandable but poorly structured, written or presented. 0: Unacceptable/incomplete report.
10. GitHub repository & reproducibility	2	2: Repository follows required structure; clear README; requirements/dependencies; organized code; reproducible instructions; data appropriately referenced rather than improperly uploaded. 1: Repository present but incomplete or difficult to reproduce. 0: Missing or unusable repository.
11. Video demonstration	2	2: Clear, well-organized ≤ 10-minute demonstration covering implementation, results and key insights. 1: Basic demonstration with omissions or weak communication. 0: Missing/unusable.
TOTAL	32	

Please reach out if you have any questions.

All the best!

